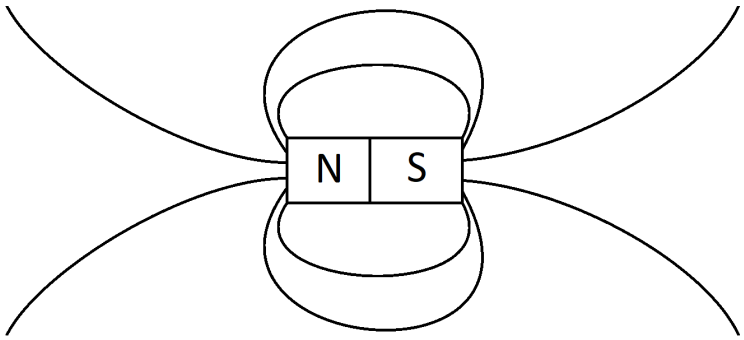
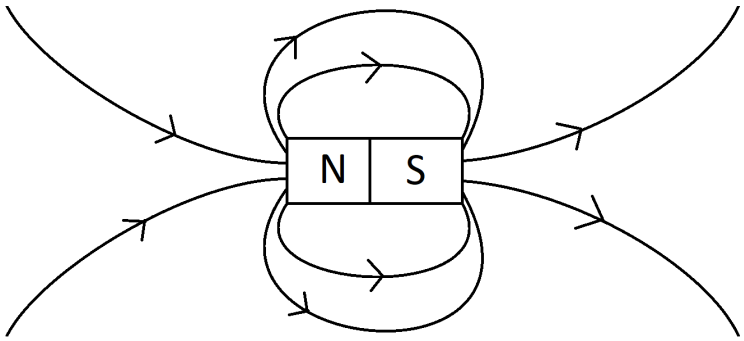
Worksheet - Electric force and electric field

Understanding the magnetic field of one magnet:

- In this picture we have one magnet with the field lines drawn. I have not drawn the arrows. Do the arrows point from North to South or from South to North?



Solution: The arrows always point from North to South on magnets.



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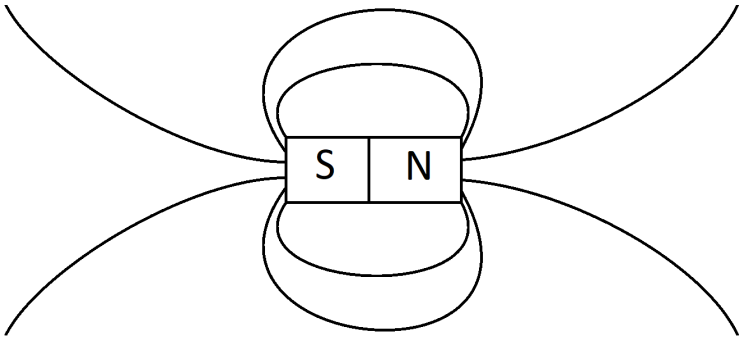

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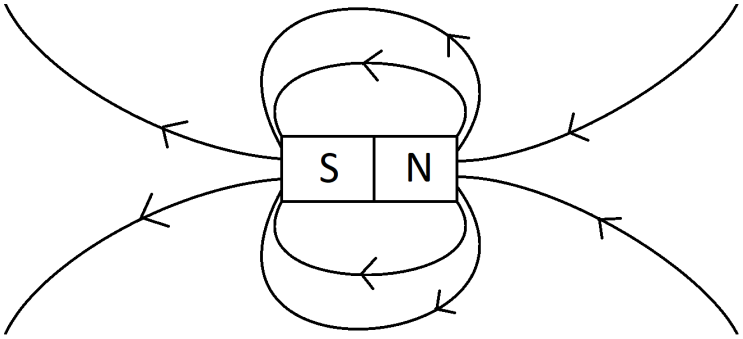
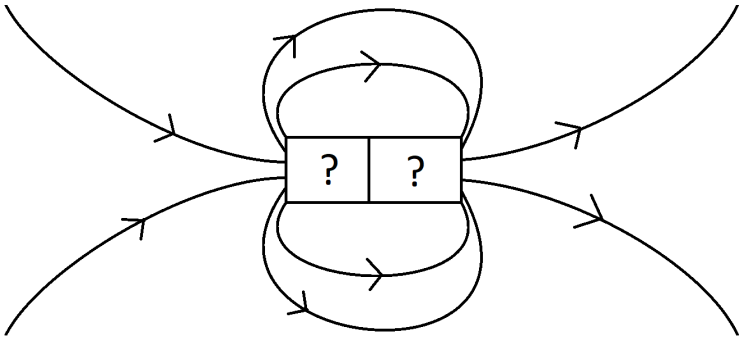
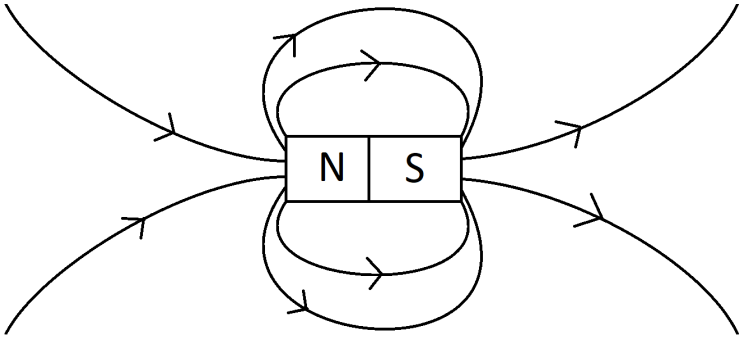


- In this picture we have one magnet with the field lines drawn. I have drawn the arrows. Is the magnet oriented North-South or South-North?

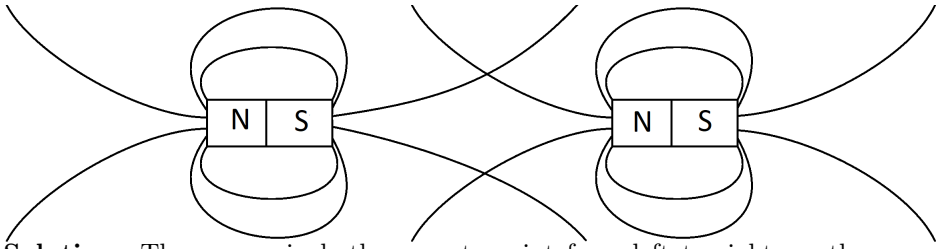


Solution: The arrows are pointing from left to right. The arrows on magnets always point from North to South so the left side must be the North pole of the magnet and the right side must be the South pole of the magnet.

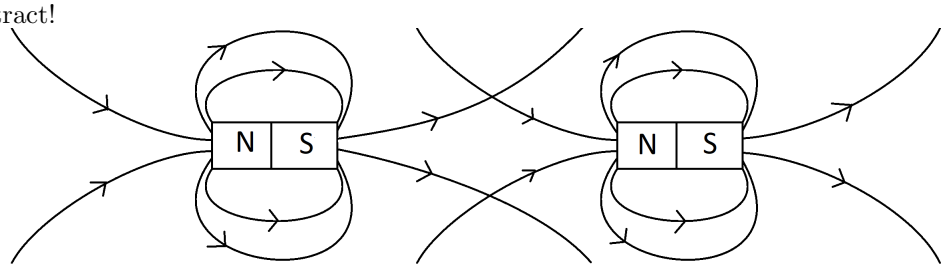


Understanding the magnetic fields of two charges:

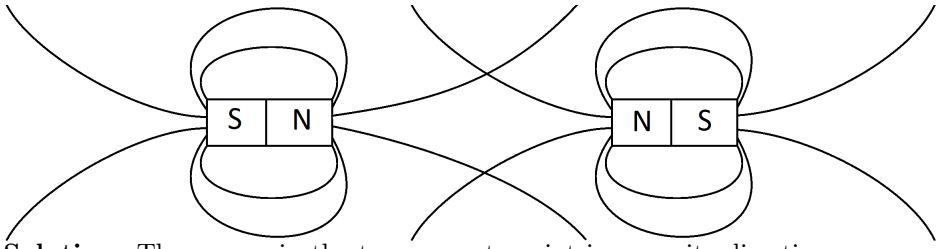
- In this picture we have two bar magnets with the field lines drawn. I have not drawn the arrows. Think about what direction the arrows go for each bar magnet. Do you expect them to attract or repel?



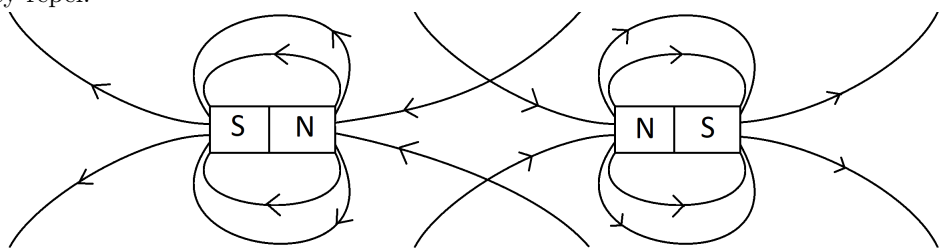
Solution: The arrows in both magnets point from left to right, so they attract!



- In this picture we have two bar magnets with the field lines drawn. I have not drawn the arrows. Think about what direction the arrows go for each bar magnet. Do you expect them to attract or repel?

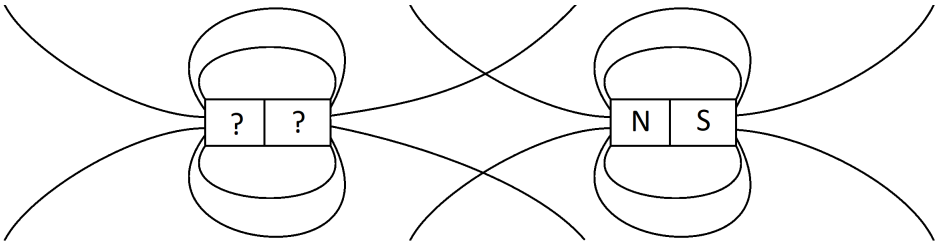


Solution: The arrows in the two magnets point in opposite directions, so they repel!

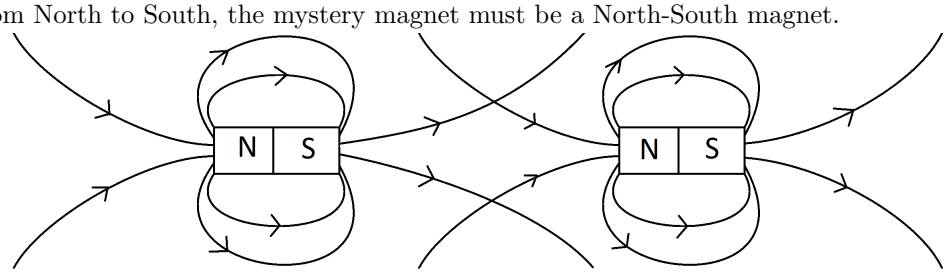


Mystery magnets:

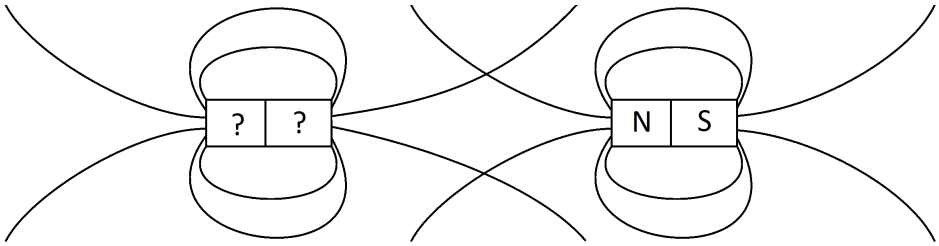
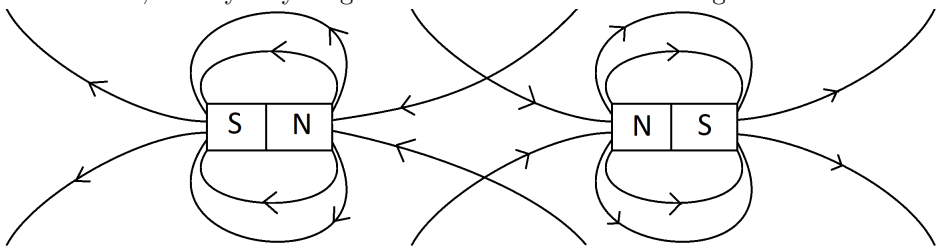
- In this picture we have a mystery magnet and a bar magnet oriented North-South from left to right with the field lines drawn. I have not drawn the arrows. The magnets attract. Is the mystery magnet oriented North-South or South-North?



Solution: The arrows of the right magnet point from left to right because it is a North-South magnet. The magnets attract, which means the arrows of both magnets point in the same direction. So the arrows of the mystery magnet must point from left to right as well. Since the arrows on a magnet always point from North to South, the mystery magnet must be a North-South magnet.



- In this picture we have a mystery magnet and a bar magnet oriented North-South from left to right with the field lines drawn. I have not drawn the arrows. The magnets repel. Is the mystery magnet oriented North-South or South-North?



Solution: The arrows of the right magnet point from left to right because it is a North-South magnet. The magnets repel, which means the arrows of the magnets point in opposite directions. So the arrows of the mystery magnet must point from right to left. Since the arrows on a magnet always point from North to South, the mystery magnet must be a South-North magnet.